Academic Resume

Oct 10, 2026 · @a

Amaara Subramaniam

Aditya Birla World Academy (ABWA)

High-school researcher in computational biology and human genetics, with a published co-authored paper, research placements at IIT Bombay and IIT, and an independent genomics platform in development.

Education

Aditya Birla World Academy (ABWA), Mumbai · Grades 9–11, graduating [2027]

- Grade 9: English (FLE), Computer Science, Biology, Spanish
- Grade 10 (International): Mathematics, Spanish, Computer Science, Economics, Biology, Physics, Chemistry, Global Perspectives
- Grade 11: English General Paper (EGP), Biology, Physics, Mathematics
- Academic Excellence awards in the subjects above, 2024–2026; General Proficiency, Grades 9–11

Research Experience

Founder & Lead Researcher, GeneMap · Independent capstone · Jan 2026 – present

Mentors: Louis (Oxford); Dr. Mayuri Bandekar (IIT Bombay)

- Building an interactive digital atlas of genetic leukodystrophies that consolidates fragmented genetic, molecular, diagnostic, biomarker and therapeutic data into one searchable platform
- Authored literature-based dossiers on eight disorders: Canavan, Krabbe, metachromatic leukodystrophy, X-linked adrenoleukodystrophy, Pelizaeus–Merzbacher, vanishing white matter, Alexander, and POLR3-related leukodystrophy
- Each dossier links primary literature, biomedical databases, clinical-trial records, standardized identifiers and evidence classifications across disease biology, genotype–phenotype relationships, biomarkers, diagnostics and therapeutic pipelines

Research Assistant · Dr. Mayuri Bandekar, IIT Bombay and BARC ·

- Investigating a novel gene in cancer progression
- Manuscript published to OJSS

Research Intern, Single-Cell Genomics & Computational Biology · Saket Chaudhary, IIT · June 15 – July 25

- Sole high-school intern in the research group
- Worked independently through the full Seurat workflow in R: QC, normalization, highly variable gene selection, PCA, clustering, UMAP, differential expression and cell-type annotation
- Reviewed about 300 scientific papers and built a structured Excel database to support curation of a healthy reference cell atlas
- Attended research seminars on current genomics work; findings expected to be published

Research Intern, Gene Therapy · BARC Hospital, under Dr. Rahul Checker · 4 weeks

- Studied plasmid design, bacterial transformation, PCR and gel electrophoresis, transfection, and protein expression and validation
- Covered AAV vectors, capsid engineering and oligodendrocyte-targeted delivery, applied to Canavan disease (ASPA/NAA biology) and its preclinical and clinical gene-therapy development
- Wrote a research report; letter of recommendation received from Dr. Checker

Publications & Submissions

- **Co-author**, "[Paper title]: Urine-Based Diagnosis of Canavan Disease," ER Publications, published. Mentor: Dr. Usha Dave. Oct 2025 – Jan 2026. Also submitted to the CREST Award.
- **Author**, novel-gene study in cancer progression (IIT Bombay / BARC; mentor Dr. Mayuri Bandekar), published to OJSS.
- **Contributor**, single-cell reference atlas project (IIT; mentor Saket Chaudhary), publication done.

Applied Projects

Founder & AI Developer, MannPakad · AI therapy app for teens with OCD

- Built an app that delivers clinically grounded OCD therapy to teens who cannot access a therapist, adapting to each user
- Clinical validation in progress with a psychologist, Dr. Nikita, and the Muskaan Foundation
- Qualified the first round of INSEF; advanced to INSEF Regionals (Grade 11)

Clinical & Laboratory Training

Genomics Intern · Dr. Usha Dave, Labassure Advanced Genomics Institute · Grade 11 · 10 sessions, 45 hours

- Trained in genetic disorder classification, chromatography, GC-MS metabolic testing, pedigree analysis, cancer biology, immunology and sample handling
- Completed laboratory biosafety training, one-to-one forensic testing training, and visits to the ACTREC cancer research laboratory (Kharghar) and biotechnology start-ups in Pune
- Wrote a final internship report; the placement led to the Canavan diagnostics research above

Genetics Observer · Dr. Hema Purandare, MedGenome / Centre for Genetic Health Care · June – July 23, 2025 · 20 hours

- Shadowed a pioneer of prenatal genetics across 6+ live counseling sessions on how DNA testing informs decisions for families

Honors & Awards

Research and national

- CREST Gold Award 2026
- INSEF Regionals, Grade 11
- Immerse Education Essay Competition, Honours List, 2026
- DELE Spanish A2, 93%, Grade 11
- Dr. Homi Bhabha Balvaidyanik Competition, participant, 2023–24
- Math Olympiad, participant, 2022–23

World Scholar's Cup, Bangkok Global Round (Aug 21, 2024)

Award	Event	Result
Gold	Overall Challenge	6th of 500 participants
Gold	Challenge: Special Area	11th
Gold	Challenge: Art & Music	15th
Gold	Collaborative Writing	50th
Gold	Scholar's Bowl, team	41st
Silver	Senior Team	10th

Award	Event	Result
Silver	South Asia Team	5th
Silver	Overall Scholar	99th
Trophy	Team Writing	6th

World Scholar's Cup, Mumbai Regional Round

- Champion Challenge Scholar (Jac Khor Trophy); Overall Scholar trophy; top-scoring scholar from school (gold)
- Team: 3rd Overall Team, Champion Challenge Team, Writing Team and Bowl Team trophies; Debate Team silver (7th)
- Gold medals: Writer, Challenge Art & Music, Challenge History, Challenge Special Area; silver medals: Debater (19th), Challenge Social Studies (5th)

School honors, ABWA

- Academic Excellence, 2024–2026 (subjects listed under Education)
- General Proficiency, Grades 9–11
- Attribute Awards, 2023 (Grade 9) and 2024, for empathy, diligence, resilience, honesty, respect and responsibility
- Muskaan Foundation Award for Social Service, Grade 11

Leadership, Service & Activities

Founder & Program Lead, Animal-Assisted Therapy · Yes I Can Foundation × Animal Angels Foundation India

- Brokered a first-of-its-kind NGO partnership bringing therapy animals to isolated elderly residents
- Certified as an animal-assisted therapist; ran the program to support residents' mental wellbeing [add sessions / residents reached]

Rescue Volunteer & Content Creator · YODA (Youth Organisation for Defence of Animals) and WSD Initiative

- Rescued and rehabilitated injured strays; ran social campaigns that drove adoptions and donations [add numbers]

Core Writer, Newsletter Team · ABWA School Newsletter · Grade 10 – present

- Published 20+ articles; started the newsletter's STEM column on current science, research by students and ethics in STEM

- Competed in the Derek O'Brien parliamentary writing contest at national level

Team Captain · World Scholar's Cup, Bangkok Global Round

Vocalist · Trinity College London

- Distinction in Grades 1 and 2; solo and band performances for 3+ years, including a 150-person concert

Trainee · Krav Maga India · Grade 9 – present

Team Member · Binary Bolts, FIRST Robotics Competition · 2022–23

- Helped design and build a competition robot under deadline

Coursework & Skills

- **Independent coursework:** Stanford Online, RNA Biology; University of Chicago, Neurobiology (both via Coursera, 90%+, no credit)
- **Computational:** R and the Seurat workflow for single-cell RNA-seq; systematic literature review and database curation
- **Laboratory exposure:** PCR, gel electrophoresis, plasmid construction, GC-MS, chromatography, sample preparation
- **Domain knowledge:** Leukodystrophies, Canavan disease, AAV gene therapy, prenatal and cancer genetics
- **Languages:** English; Spanish (DELE A2, 93%)